Hardware for a one-building pilot

Everything needed to monitor the power source and voltage, measure the tank and switch the borehole pump. All parts are sold by Nigerian electronics sellers (Computer Village, Jumia, Hub360). A.D.A. configures everything; mains work is done by a qualified electrician.

ITEM	QTY	PRICE (₦)	SOURCE
ESP32 development board (WROOM-32) The controller: Wi-Fi, reports to the hub, drives the relays	1	₦15,000	Hub360 listing
5 V power supply (HLK-PM01 module or 5 V 2 A adapter) Powers the controller from mains	1	₦2,500 – 5,000	estimate
4-channel 5 V relay module (optocoupled) Low-voltage switching signals; drives the contactor coil	1	₦5,100 – 7,500	Hub360 / Jumia
Contactor, 25 A, 2-pole, 230 V coil Switches the pump motor safely (one per heavy load); fitted by electrician	1	₦15,000 – 35,000	estimate
PZEM-004T v3 energy meter (100 A, with CT) Isolated mains voltage, current, power and energy	1	₦19,000 – 25,000	Microscale listing
220 V AC detection module (optocoupler) Tells NEPA from generator: one on each supply	2	₦3,000 – 5,000 ea.	estimate
JSN-SR04T waterproof ultrasonic sensor Overhead tank level	1	₦5,700 – 10,000	Microscale listing
IP65 ABS enclosure (about 200 × 150 × 75 mm) Houses the controller near the distribution board	1	₦8,000 – 15,000	estimate
DIN rail, terminal blocks, cable glands Neat, safe terminations	1 set	₦5,000 – 10,000	estimate
6 A MCB or fused spur for the controller supply Protects the controller circuit	1	₦4,000 – 8,000	estimate
Signal cable for tank sensor (alarm/CAT5, 20–30 m) Tank on the roof to the controller	1 run	₦6,000 – 15,000	estimate
1.5 mm² cable, jumpers, sundries Internal wiring	—	₦5,000 – 10,000	estimate
Core parts subtotal		≈ ₦96,000 – 166,000	

Optional, depending on the site: always-on hub device, e.g. Raspberry Pi or spare Android phone (₦0 – 120,000; a laptop or phone can run the hub during the pilot) · Wi-Fi router or MiFi if the building has none (₦20,000 – 40,000) · spare ESP32 board (₦15,000).

Core parts
₦96k – 166k
controller, sensing, switching, enclosure

Electrician (installation)
₦20k – 50k
estimate; varies by site and scope

Expected pilot total
≈ ₦116k – 216k
before optional items

About these prices. Named sources are Nigerian seller listings seen in November 2026 (some were out of stock); "estimate" means a typical market range to confirm with a supplier. Each extra switched load (e.g. an AC) adds one contactor. Who buys and funds the parts is agreed before installation.

Ajala David · +234 916 107 8549